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For this project, I decided to build a SQL database for an Aquarium & Fish Care Store mini world using Monster Aquarium Inc in Queens, New York as my real-world example. I picked this scenario because it felt more meaningful than choosing a random business idea. I actually enjoy going to different pet stores in the city, and Monster Aquarium is one of my favorite places to visit when I want to buy fish or supplies for my tanks. I currently take care of three fish tanks at home, so I already understand how important it is to keep track of things like food, filters, water conditioners, and compatible fish. Because of that, using an aquarium store made the assignment feel realistic and easier for me to connect to.

To make the system design more accurate, I went to Monster Aquarium in person and asked the owner a few questions. The owner is also the manager of the store, so he was able to explain how the store works from both a business and daily operations point of view. I asked him how they manage inventory, how they decide what to restock, and what information they usually need when customers buy items. He explained that they work with suppliers that provide fish, aquarium products, and supplies, and that the store constantly needs to track which items are in stock and which items are running low to be able to re resupply before running out of stock. He also said that customers often ask questions about fish care, and employees help them pick products based on the customer’s needs. From our conversation, I learned that a real aquarium store is not just about selling products, but it also involves managing inventory carefully and helping customers make good choices, so the fish stay healthy.

Based on the requirements I gathered, I decided my database should focus on the main parts of how the store runs. The entities I chose represent the key things the store needs to track. For example, the Product entity is important because the store sells many different kinds of items, and each product needs details like a name, price, and whether it is in stock. The Supplier entity is needed because products come from different vendors, and the store needs to know where items are coming from for restocking. I also included a Customer entity because customers are the ones placing orders, and it is useful to store basic customer details like name and contact information. The Orders entity records each transaction, which is important for keeping a purchase history. To make the ordering process more realistic, I included an OrderItem entity, since an order can include multiple items and order_item acts as the bridge between orders and products. Finally, I used a Category entity to group products into types (like fish, food, filtration, water treatment, plants, etc.) so inventory is easier to organize. Overall, this project helped me understand how databases connect to real businesses, and visiting Monster Aquarium made the design process feel more like building an actual system that that I can use one day instead of just doing an assignment.